

EGM 5421: Structural Dynamics - Project Report

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1 Question

A cellular telecommunication tower is to be analyzed for a variety of dynamic loads.

1.1 Part: A

First to identify its dynamic properties modeled as a SDOF, a lateral force of 10000 lb is applied at the top of the tower resulting in a horizontal displacement of 1 inch. A cable attached to the top of the tower is suddenly severed resulting in a free vibration. At the end of 20 complete vibration cycles, the time is 200 sec and the amplitude is 0.02 inch. Find effective stiffness, effective mass, undamped natural frequency and damping ratio for the SDOF system.

Solution:

$$k = \frac{F}{u} = \frac{10000}{1} = 10000 \text{ lb/inch} = 10 \text{ kips/inch},$$

$$T_n = \frac{200}{20} = 10 \text{ sec},$$

$$w_n = \frac{2\pi}{T_n} = \sqrt{\frac{k}{m}} = 0.628 \frac{\text{rad}}{\text{sec}} = \sqrt{\frac{10}{m}}$$

$$m = 25.256 \frac{\text{kips sec}^2}{\text{inch}}$$

$$W = 25.356 \times 386 = 9789.95 \text{ kips}$$

$$\xi = \frac{1}{2\pi j} \ln\left(\frac{u_i}{u_{i+j}}\right) = \frac{1}{2\pi \times 20} \ln\left(\frac{1}{0.02}\right) = 3.11\%.$$

1.2 Part: B

The tower must survive an extreme natural event, such that the imposed loading can be modeled as a superposition of a harmonic, triangular and square wave loading whose characteristics are given as follows:

- 1) harmonic loading – amplitude= 10 kips, period=0.1 s and duration=5 s,
- 2) triangular loading – amplitude= 15 kips, period=0.12 s and duration=4 s,
- 3) square loading – amplitude= 5 kips, period=0.15 s and duration=5 s.

Using Nonlin, find the imposed wave functions and its FFT. Find the response of the system assuming that the system is linear. Is the tower designed sufficiently well if the design criterion is a maximum deflection of 0.5 inch.

Results:

STRUCTURE PROPERTIES INPUT

Unit Type: ☒ U.S. ☐ METRIC

Length Units: ☒ inches ☐ feet

Force Units: ☐ pounds ☒ kips

Dynamic Force Applied As...: ☐ Free Vibration ☒ Forcing Function ☐ Blast Load ☐ Ground Acceleration ☐ Inc. Dynamic Analysis

Constitutive Properties: ☒ 10.00 k/in ☐ STIFFNESS K2 ☐ STRENGTH ☐ STIFFNESS KG

☒ LINEAR ANALYSIS ☐ NONLINEAR ANALYSIS ☐ INCLUDE P-DELTA

Dynamic Properties:

Period, seconds	10.005
Frequency, Hertz	0.100
Frequency, R/sec	0.628
Effective K1, k/in	
Effective Fy, k	
Damping	0.990 k-sec/in
Mass	25.356 k-sec ² /in
Gravity	386.1 in/sec/sec

Summary of Latest Run:

Spring Force	
Damping Force	
Spring + Damping	
Displacement	
Ductility Demand	
Yield Events	
Energy	

9789.95 k

3.11 % critical

FORCING FUNCTION

Figure 1: Nonlin interface and input data.

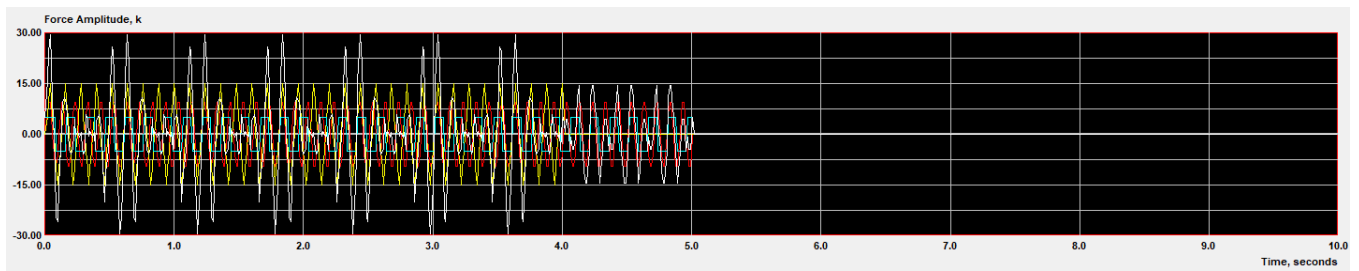


Figure 2: Imposed wave functions.

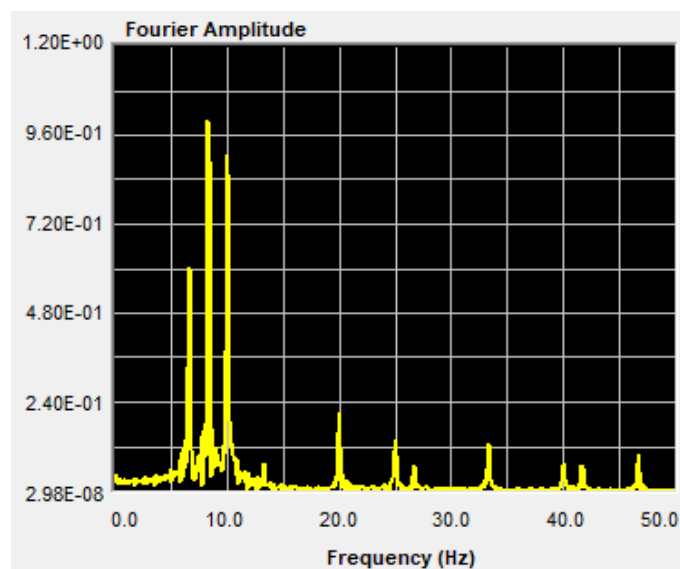


Figure 3: Fast Fourier Transform.

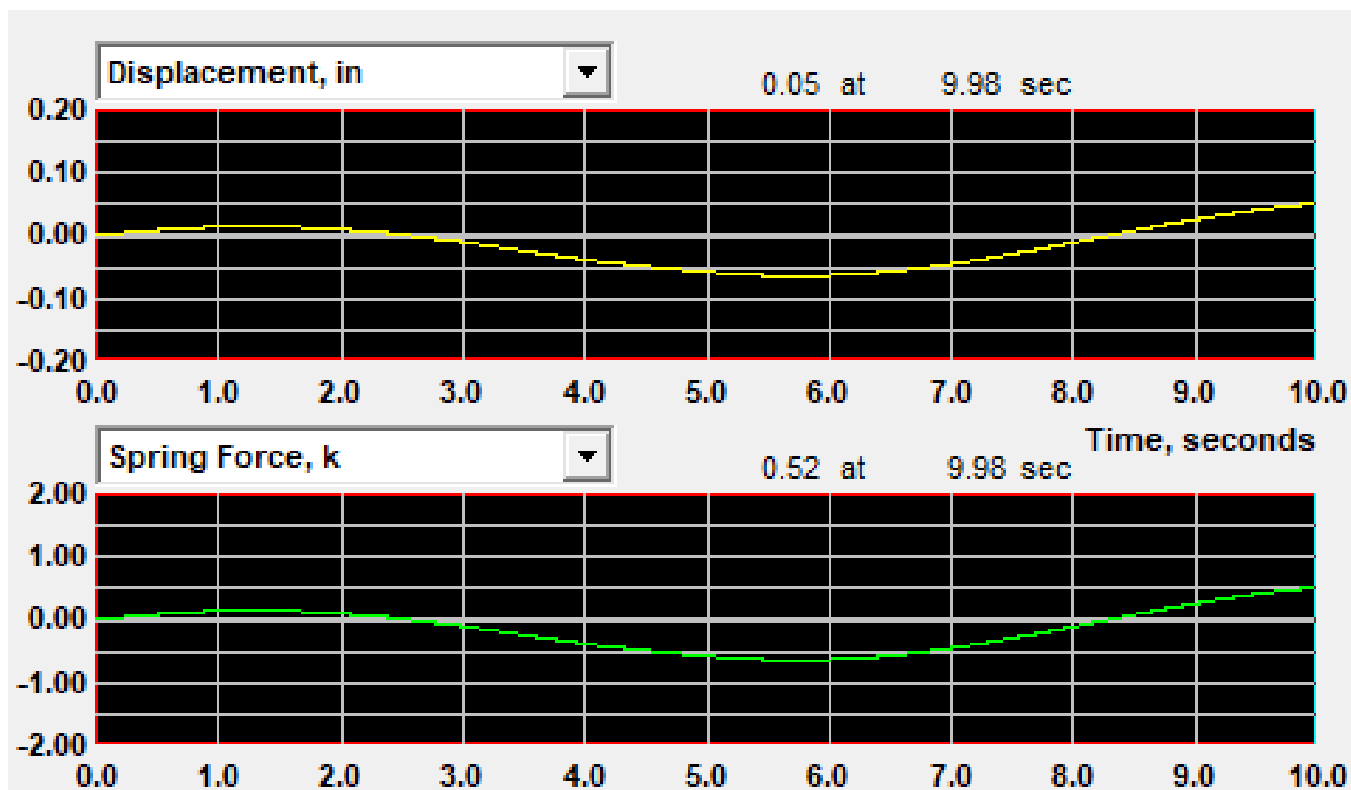


Figure 4: Displacement and spring force versus time.

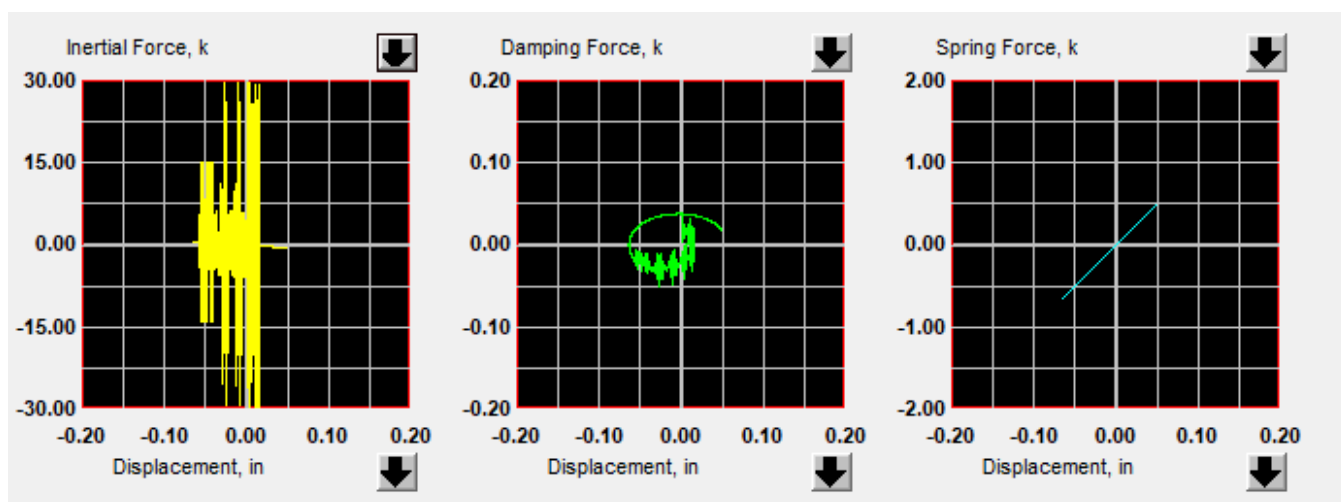


Figure 5: Inertial force, damping force and spring force versus displacement.

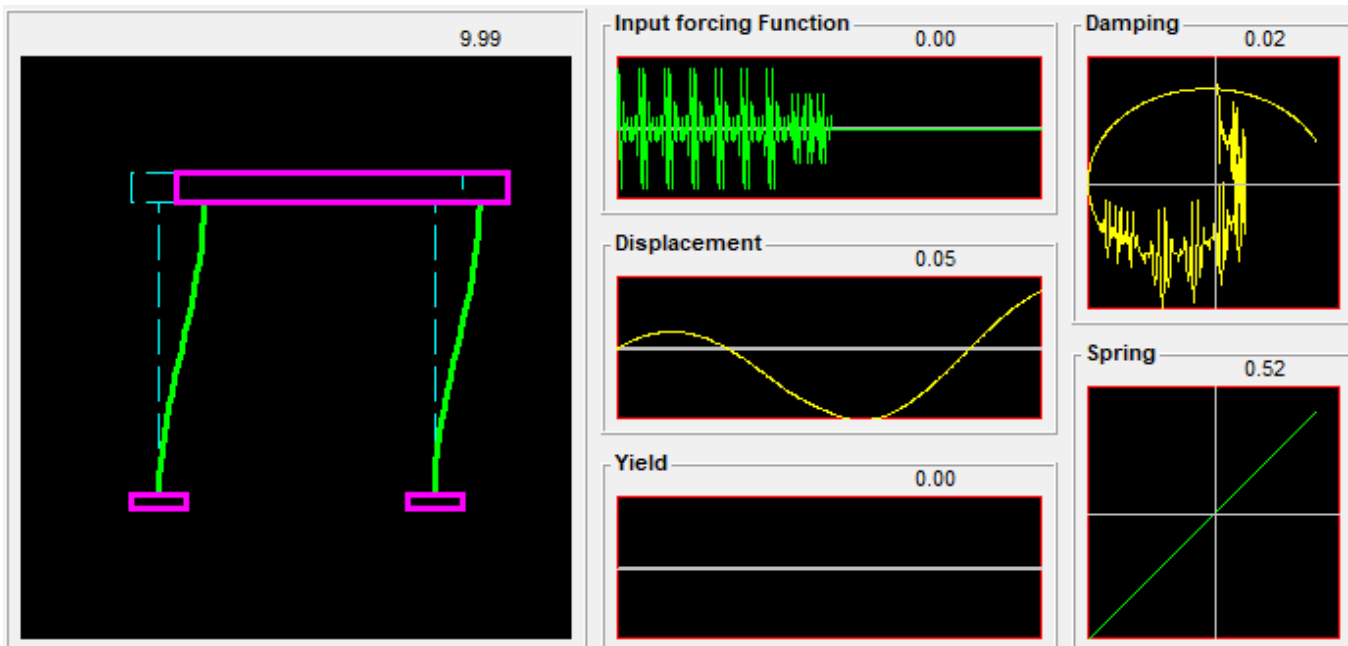


Figure 6: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 29.8192 k
- Maximum Damping Force 0.0506 k
- Maximum Spring Force 0.6394 k
- Maximum Damping + Spring Force 0.6406 k
- Maximum Total Force (D+S+I) 29.5108 k
- **Maximum Computed Displacement 0.0639 in**
- Maximum Computed Velocity 0.0511 in/sec
- Maximum Computed Acceleration 1.1760 in /sec²

The tower is adequately designed, as the maximum deflection observed (0.0639 inches) is well within the design criterion of 0.5 inches.

1.3 Part: C

If the tower is not sufficiently well designed, then suggest alternatives to the owner.

Answer:

The design of the tower should be examined and improved to guarantee it can support dynamic loads without going over deflection limitations. The proposed changes—such as raising stiffness, including damping devices, and strengthening the base or structural components—depending on the outcomes of future studies or inspections could serve to strengthen the tower. Regular assessments will also guarantee the tower keeps performance requirements and safety criteria.

Increasing Stiffness: The stiffness of the system is doubled.

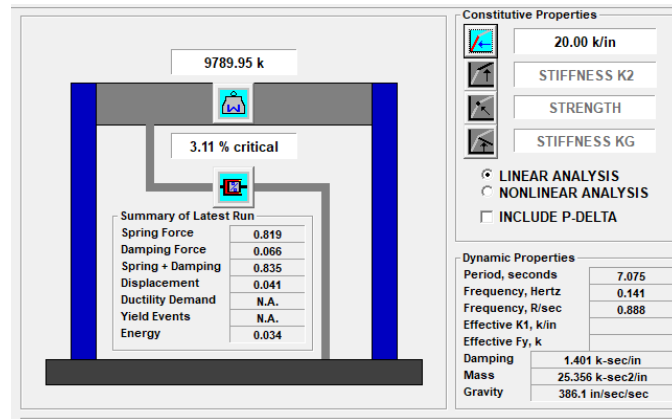


Figure 7: Nonlin interface and input data for increment in stiffness.

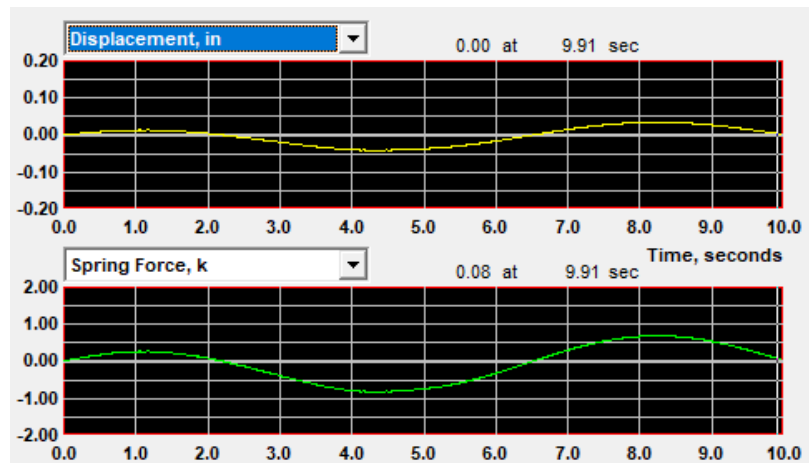


Figure 8: Displacement and spring force versus time.

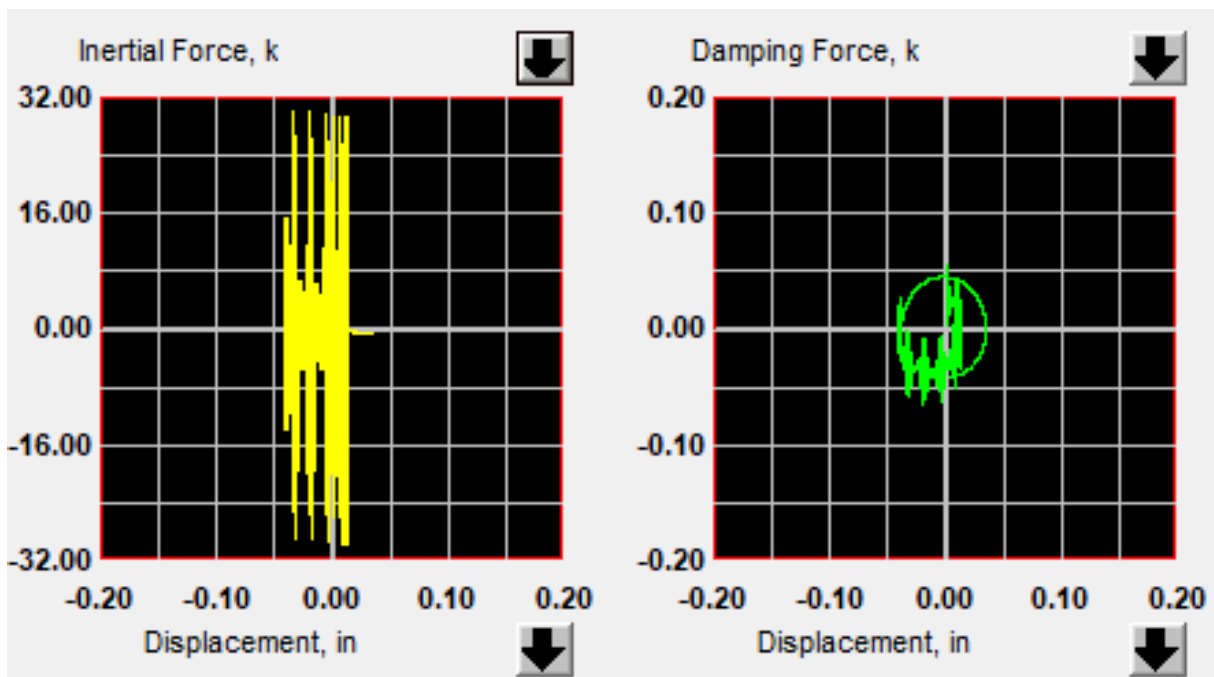


Figure 9: Inertial force and damping force versus displacement.

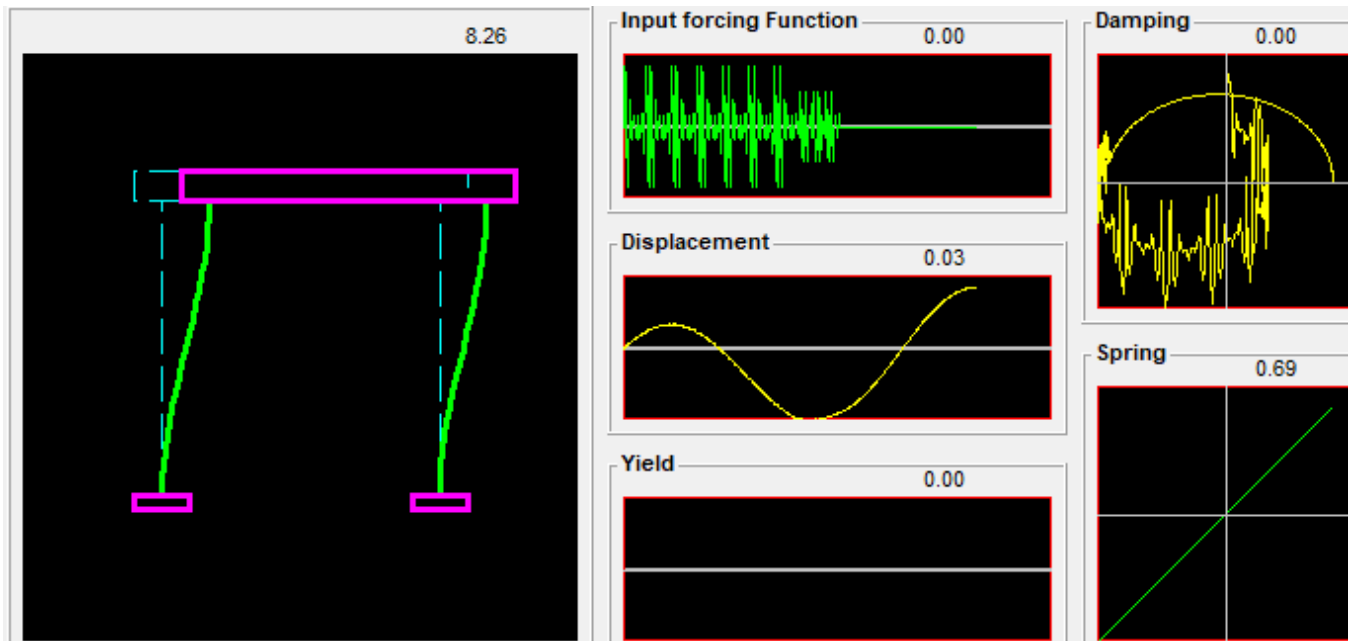


Figure 10: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 30.2178 k
- Maximum Damping Force 0.0660 k
- Maximum Spring Force 0.8190 k
- Maximum Damping + Spring Force 0.8351 k
- Maximum Total Force (D+S+I) 29.5108 k
- **Maximum Computed Displacement 0.0410 in**
- Maximum Computed Velocity 0.0471 in/sec
- Maximum Computed Acceleration 1.1917 in /sec²

When the system stiffness was doubled, the observed maximum deflection decreased to 0.0410 inches. However, The maximum inertial force, damping force and spring force were increased due to the increment in stiffness.

Increasing Mass: The mass of the system is nearly doubled.

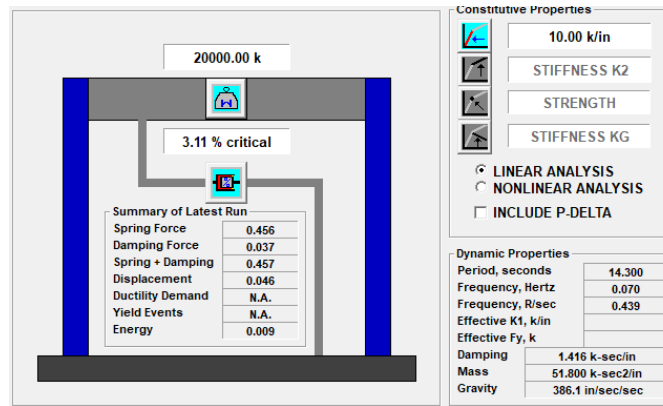


Figure 11: Nonlin interface and input data for increment in mass.

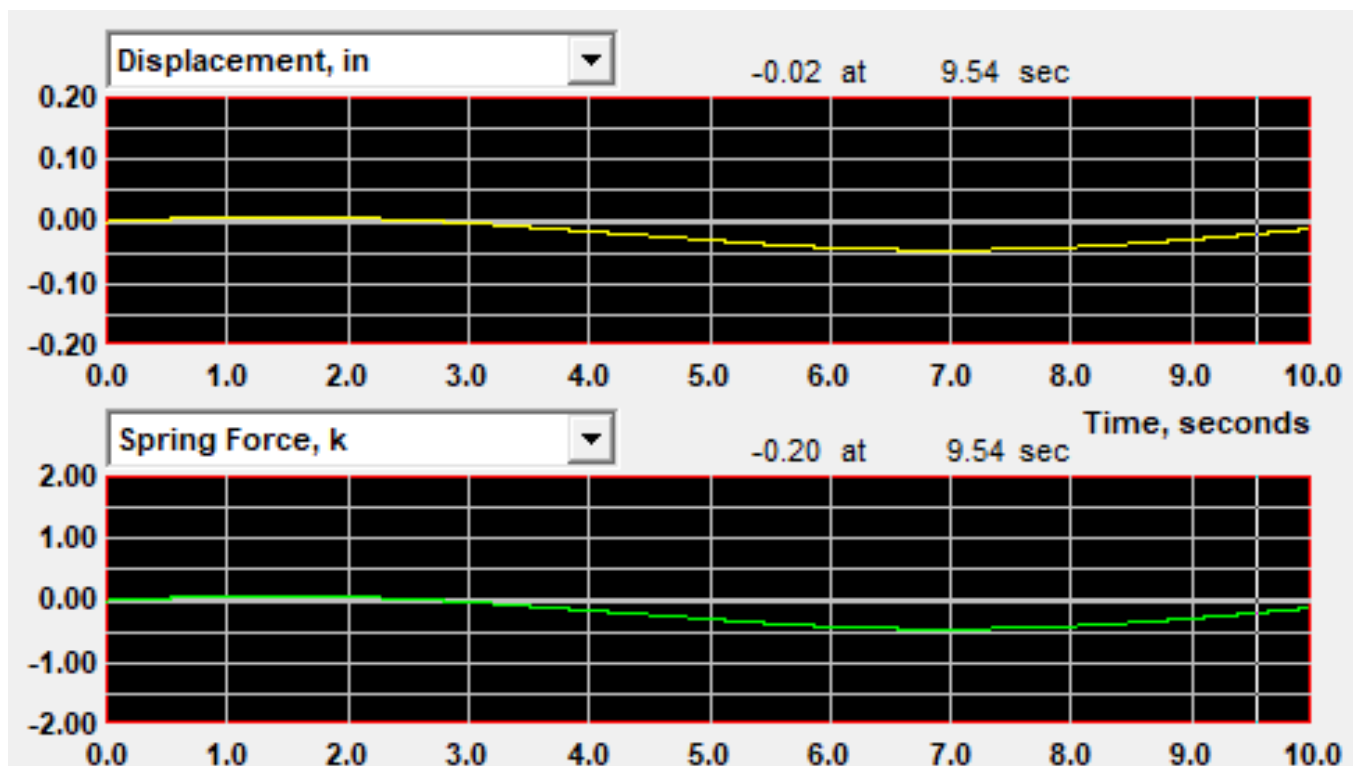


Figure 12: Displacement and spring force versus time.

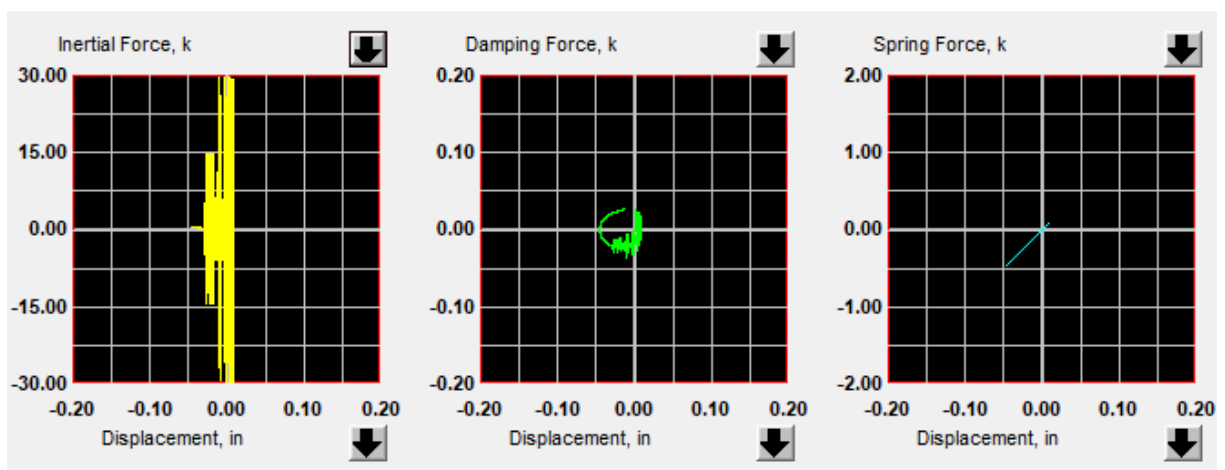


Figure 13: Inertial force, damping force and spring force versus displacement.

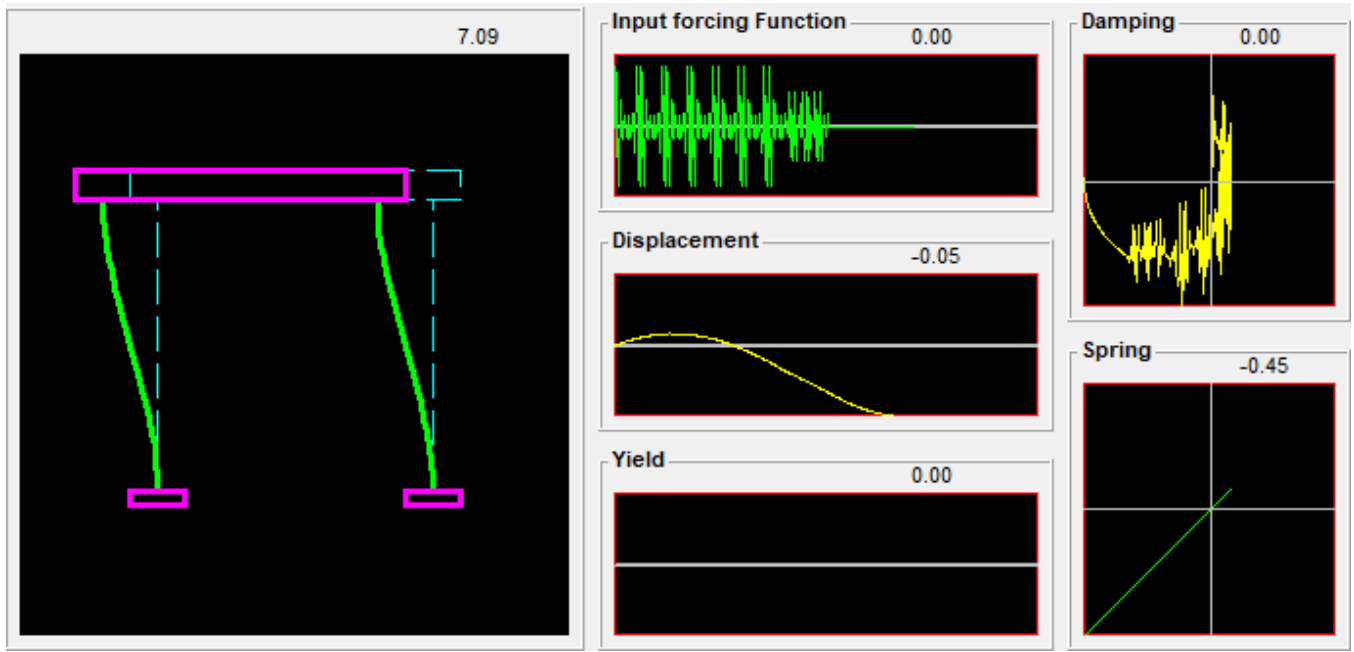


Figure 14: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 29.6383 k
- Maximum Damping Force 0.0366 k
- Maximum Spring Force 0.4558 k
- Maximum Damping + Spring Force 0.4567 k
- Maximum Total Force (D+S+I) 29.5108 k
- **Maximum Computed Displacement 0.0456 in**
- Maximum Computed Velocity 0.0259 in/sec
- Maximum Computed Acceleration 0.5722 in /sec²

When the system mass was nearly doubled, the observed maximum deflection decreased to 0.0456 inches. Further, the maximum velocity and acceleration were reduced to approximately half of their original values.

1.4 Part: D

What will be the response of the original system if the tower has a yield point is 0.5 kips and the secondary stiffness is 1 kip/in. What is the permanent deflection of the tower. Will the tower survive this event.

Results:

The screenshot shows the 'STRUCTURE PROPERTIES INPUT' dialog box. It contains several sections for configuring the analysis:

- Unit Type:** ☒ U.S., ☐ METRIC
- Length Units:** ☒ inches, ☐ feet
- Force Units:** ☐ pounds, ☒ kips
- Dynamic Force Applied As...:** ☐ Free Vibration, ☒ Forcing Function, ☐ Blast Load, ☐ Ground Acceleration, ☐ Inc. Dynamic Analysis
- Constitutive Properties:**
 - 10.00 k/in
 - 1.00 k/in
 - 0.50 k
 - STIFFNESS KG
- Analysis Type:** ☐ LINEAR ANALYSIS, ☒ NONLINEAR ANALYSIS, ☐ INCLUDE P-DELTA
- Dynamic Properties:**

Period, seconds	10.005
Frequency, Hertz	0.100
Frequency, R/sec	0.628
Effective K1, k/in	
Effective Fy, k	
Damping	0.990 k-sec/in
Mass	25.356 k-sec ² /in
Gravity	386.1 in/sec/sec
- Summary of Latest Run:**

Spring Force	0.516
Damping Force	0.051
Spring + Damping	0.534
Displacement	0.066
Ductility Demand	1.314
Yield Events	1
Energy	0.034

The central part of the dialog shows a schematic of a tower structure with a horizontal beam at the top. Above the beam, the text '9789.95 k' is displayed. Below the beam, the text '3.11 % critical' is shown. The tower is supported by a base, and the entire structure is rendered in blue and grey.

Figure 15: Nonlin interface and input data for nonlinear analysis.

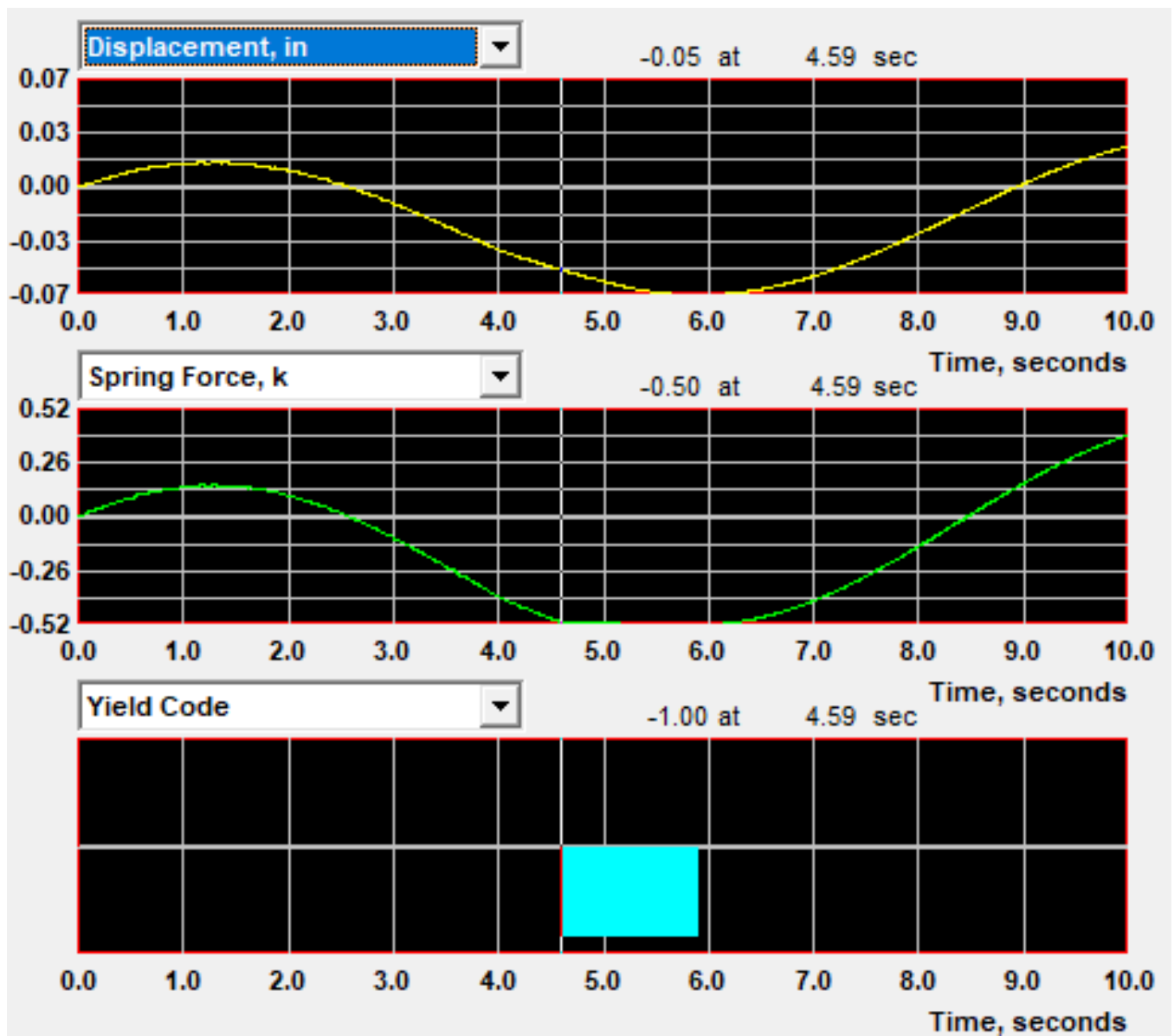


Figure 16: Displacement, spring force and yield versus time.

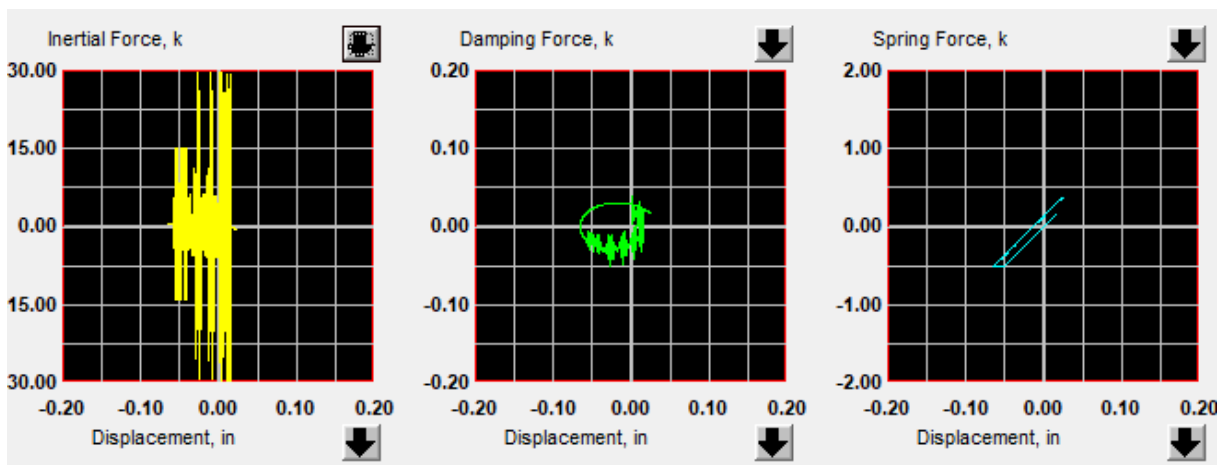


Figure 17: Inertial force, damping force and spring force versus displacement.

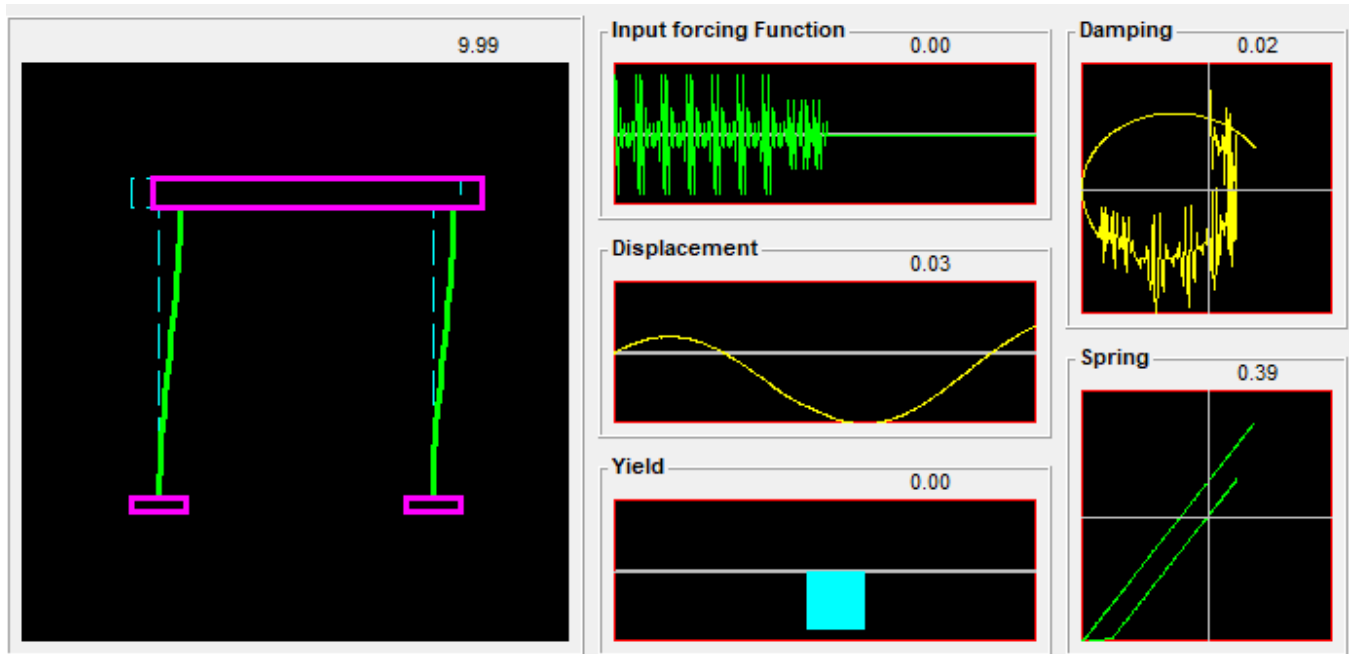


Figure 18: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 29.8192 k
- Maximum Damping Force 0.0506 k
- Maximum Spring Force 0.5157 k
- Maximum Damping + Spring Force 0.5335 k
- Maximum Total Force (D+S+I) 29.5108 k
- **Maximum Computed Displacement 0.0657 in**
- Maximum Computed Velocity 0.0511 in/sec
- Maximum Computed Acceleration 1.1760 in /sec²
- Number of Yield Excursions 1
- Displacement Ductility Demand 1.3141

The permanent deflection (plastic displacement) is calculated to be approximately $0.07 - 0.045 = 0.025$ inches based on the spring force plot in Fig. 17. As the maximum deflection of 0.0657 inches remains below the allowable threshold of 0.5 inches, the tower successfully withstands the event.

1.5 Part: E

The tower is also subjected to an impact loading of 1000 kips of duration 0.05s. What will be the response, first assuming linear behavior and, subsequently, assuming nonlinear behavior.

Results:

Linear Analysis:

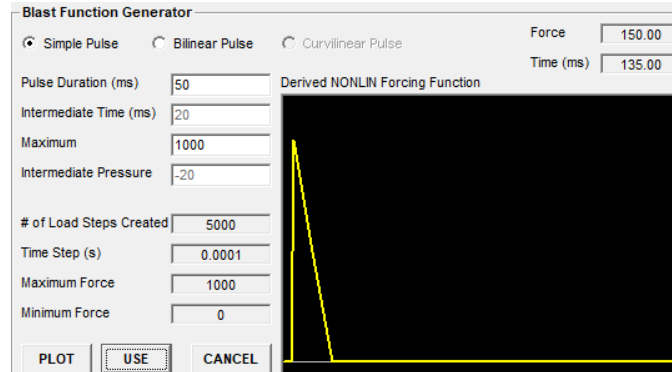


Figure 19: Blast loading.

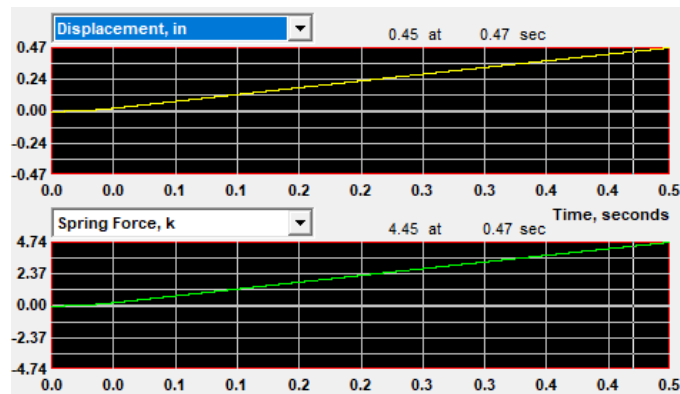


Figure 20: Displacement and spring force versus time.

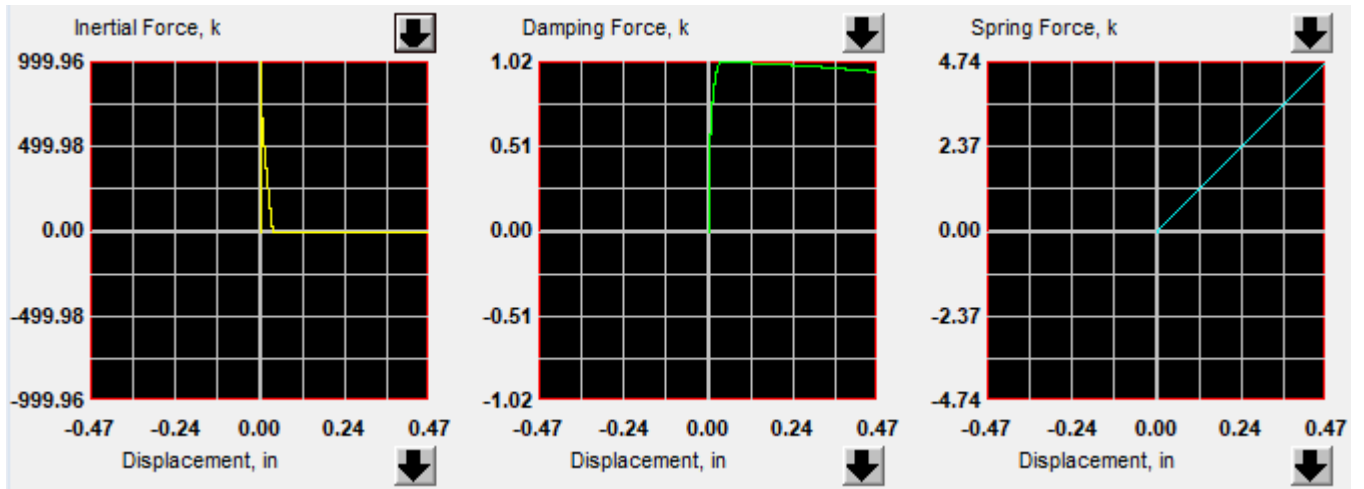


Figure 21: Inertial force, damping force and spring force versus displacement.

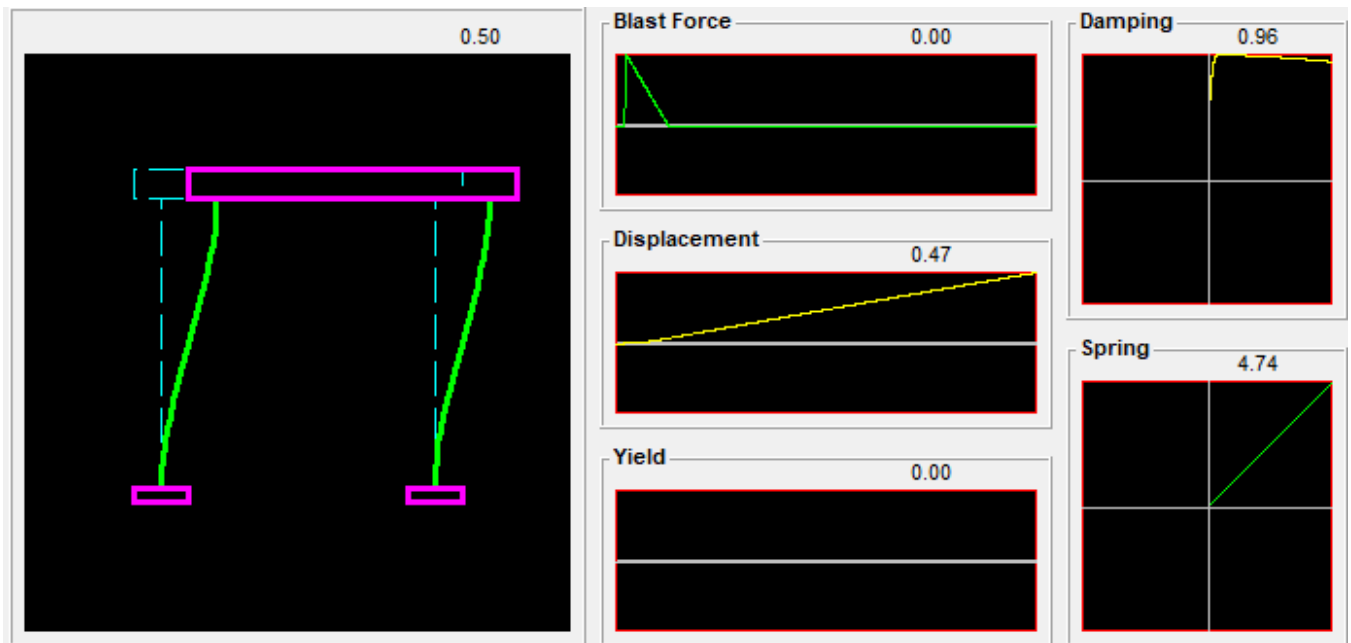


Figure 22: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 999.9007 k
- Maximum Damping Force 1.9990 k
- Maximum Spring Force 8.6706 k
- Maximum Damping + Spring Force 8.8410 k
- Maximum Total Force (D+S+I) 1000.0000 k
- **Maximum Computed Displacement 8.2206 in**
- Maximum Computed Velocity 2.0183 in/sec
- Maximum Computed Acceleration 39.4345 in /sec²
- Number of Yield Excursions 2
- Displacement Ductility Demand 164.4118

Nonlinear Analysis:

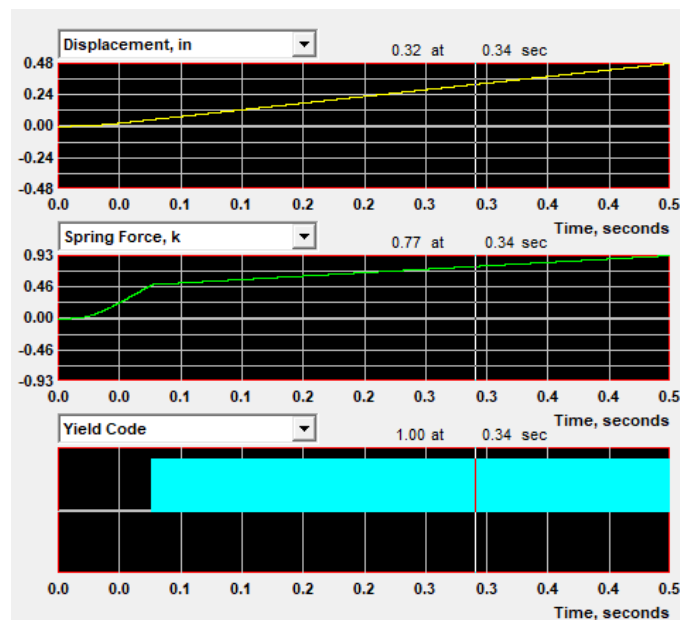


Figure 23: Displacement, spring force and yield versus time.

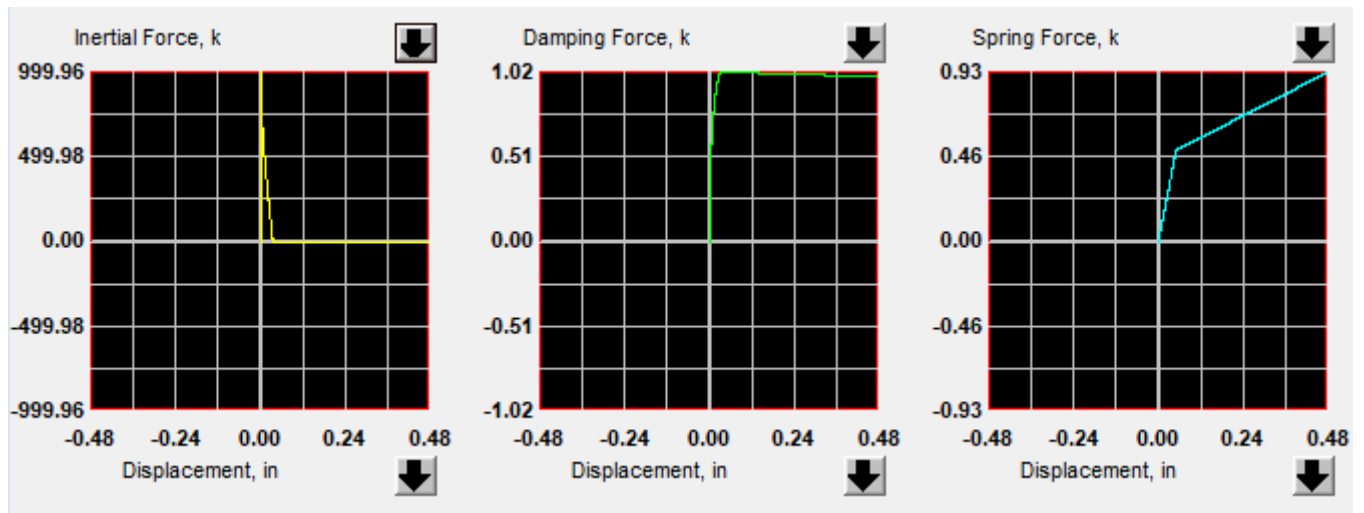


Figure 24: Inertial force, damping force and spring force versus displacement.

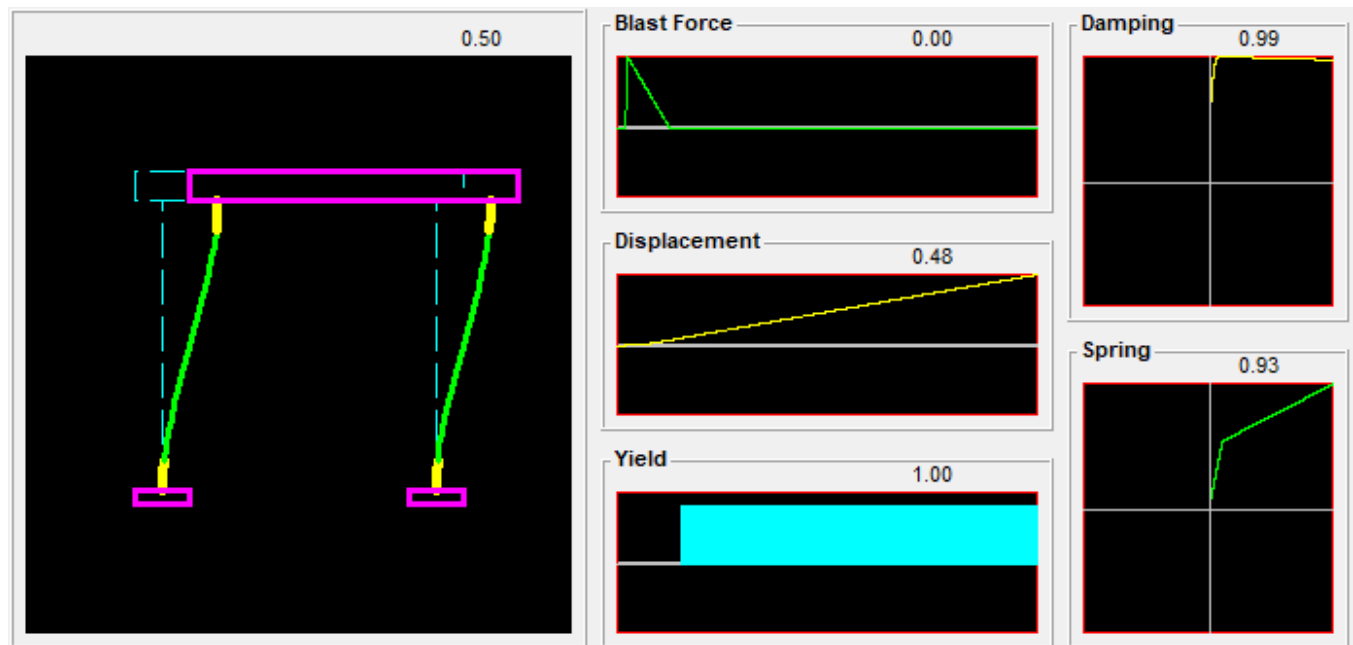


Figure 25: Animation of structure results.

SUMMARY OF RESPONSE MAXIMA:

- Maximum Inertial Force 999.9608 k
- Maximum Damping Force 1.0179 k
- Maximum Spring Force 0.9290 k
- Maximum Damping + Spring Force 1.9176 k
- Maximum Total Force (D+S+I) 1000.0000 k
- **Maximum Computed Displacement 0.4790 in**
- Maximum Computed Velocity 1.0277 in/sec
- Maximum Computed Acceleration 39.4369 in /sec²
- Number of Yield Excursions 1
- Displacement Ductility Demand 9.5794